Early Game Win Prediction in League of Legends Using Machine Learning

Annie Lee
Department of Statistics
University of Michigan
Ann Arbor, MI
minianni@umich.edu

***Abstract*—This project explores the predictive potential of early-game data in League of Legends by building machine learning models to classify match outcomes. Using a high diamond ranked dataset and various classifiers such as Logistic Regression, Random Forest, XGBoost, Support Vector Machine, and Multi-layer Perceptron, we evaluate performance through accuracy, F1 score, and ROC AUC. Our findings highlight the predictive power of early objectives and gold and experience differences. We also discuss model interpretability, propose extensions with champion pick data, and outline future improvements. This study contributes to real-time eSports analytics by bridging gameplay strategy with data science.**

Keywords—League of Legends, eSports analytics, machine learning, early-game prediction, classification models, ROC AUC, champion draft features

I. Introduction

A. Background and Motivation

As someone who has played League of Legends for over thirteen years, I've developed a deep appreciation for the game's evolving complexity and competitive richness. My long-term engagement has sparked a strong curiosity: can the outcome of a match be predicted from data alone? This project stems from that interest, combining personal experience with analytical modeling to explore win prediction in a game I've followed closely for more than a decade.

League of Legends (LoL) is a leading multiplayer online battle arena (MOBA) game with a large global player and viewer base. Its competitive depth comes from not only individual skill but also team coordination, objective control, and early-game advantages. In professional play, teams use data, replays, and predictive models for strategic insights [1].

Predicting match outcomes from early-game data is a growing focus in eSports analytics. While previous studies have used statistics and machine learning to evaluate performance, many models lack real-time relevance or overlook the strategic value of early objectives and gold distribution.

B. Project Goal

This project aims to build and evaluate predictive models that classify match outcomes (win/loss) for high-ranked League of Legends games using statistical indicators recorded in the first 10 minutes of gameplay. By comparing multiple machine learning algorithms—including Logistic Regression, Random Forest, XGBoost, SVM, and MLP—we seek to determine which models are best suited for this task and which game features are most influential in determining early-game dominance.

C. Related Work Overview

Community-driven analyses have shown correlations between early-game objectives (e.g., dragons, heralds), gold advantage, and eventual win rate [2]. This project builds on such findings by applying modern classification techniques with rigorous model selection and evaluation, including visualization and interpretability tools to provide both predictive accuracy and actionable game insights.

In terms of methodological foundation, this project draws from two seminal works in statistical learning. The first is An Introduction to Statistical Learning with Applications in Python, which emphasizes the practical implementation of machine learning methods using real-world data [3]. The second is the more theoretical The Elements of Statistical Learning, which provides in-depth mathematical foundations for classification, regression, and ensemble methods, including decision trees, SVMs, and neural networks [4].

These references provided critical guidance in selecting appropriate models, understanding trade-offs between complexity and interpretability, and designing a robust evaluation framework for this eSports prediction problem.

II. Method

The goal of this project is to predict the outcomes (win or lose) of League of Legends (LoL) matches at the high-diamond rank using match statistics collected during the first 10 minutes of gameplay. The problem is formulated as a binary classification task, where input features include various in-game metrics, and the output is a binary indicator of match outcome (1 if the blue team wins, 0 otherwise).

A. Dataset

The dataset utilized in this study is the "High Diamond Ranked 10 Minute" dataset available publicly, containing 9,879 match records with detailed statistics of early-game performance metrics, such as gold earned, experience levels, objectives taken, and player-specific metrics. The data is split into training and testing sets using stratified sampling (80%-20%) to maintain class distribution.

B. Model Formulation

1) Logistic Regression (LR): A baseline linear classifier with regularization.

2) Random Forest (RF): An ensemble model optimized using hyperparameter tuning with grid search on the depth and number of estimators.

3) XGBoost: A powerful gradient boosting framework optimized by iterative tuning to efficiently handle feature interactions and prevent overfitting.

4) Support Vector Machines (SVM): Optimized with hyperparameter tuning (kernel type and regularization parameter), this method maps input data into higher-dimensional space to find an optimal separating hyperplane that maximizes the margin between different classes.

5) Multi-layer Perceptron (MLP): A neural network model configured with optimized hidden layer architectures

and tuned hyperparameters (such as learning rate, number of epochs, and activation functions) to capture complex non-linear relationships within the data.

C. Methodology

1) Data preprocessing: Initial exploration and handling of missing values.

2) Feature Engineering: Selecting relevant features, removing identifiers, and ensuring feature relevance.

3) Model Training: Using stratified cross-validation (5-folds) for hyperparameter tuning to optimize ROC AUC scores.

4) Model Evaluation: Evaluating models using Accuracy, F1 Score, and ROC AUC metrics, providing a comprehensive performance comparison.

5) Visualization: ROC AUC curves are plotted to visualize model performance and facilitate hyperparameter selection decisions.

III. Result

Following the preprocessing steps and feature engineering described earlier, several classification models were trained and evaluated. The dataset was split into training and test sets (80%-20%) using stratified sampling to ensure balanced class distribution. Hyperparameter optimization was conducted through a 5-fold stratified cross-validation approach to maximize ROC AUC performance.

A. Numerical Simulation Results

TABLE I. The Evaluation Metrics For Each Classifier

Model	***Accuracy(%)***	***F1 Score***	***ROC AUC(%)***
Logistic Regression	71.96	0.7204	80.61
Random Forest	71.71	0.7131	79.94
XGBoost	69.73	0.6955	77.89
Support Vector Machine	71.69	00.7168	80.13
Multi-layer Perceptron	64.42	0.7266	74.29

As shown in Table I, Logistic Regression and SVM achieved comparable accuracy (71.96% and 71.71%) and ROC AUC scores (80.61% and 80.13%). However, Logistic Regression slightly outperformed SVM, indicating its higher robustness and better generalization capability. The Random Forest model also demonstrated competitive performance, closely trailing Logistic Regression and SVM, with slightly lower F1 and ROC AUC scores. XGBoost performed moderately, showing room for improvement, potentially due to overfitting or suboptimal parameter tuning. The MLP exhibited the lowest accuracy but maintained a relatively high F1 score, suggesting its sensitivity to class imbalance and difficulty in generalizing from limited features.

B. Visualization Analysis

The visualizations provided critical insights into model performance and feature importance in the context of League of Legends gameplay:

1) Logistic Regression Hyperparameter Tuning: The inverse regularization strength (C) of 1 yielded the optimal ROC AUC performance, indicating that moderate regularization best captured the relationship between early-game events (such as dragon and herald kills) and game outcomes.

2) Random Forest Hyperparameter Tuning: A max depth of 5 and 200 estimators provided the best balance, suggesting the importance of controlled complexity to effectively capture meaningful patterns like early game gold and experience differentials without overfitting.

3) ROC Curves: Logistic Regression (AUC=80.62%) and SVM (AUC=80.13%) exhibited superior performances, implying their effectiveness in accurately classifying matches based on early advantage indicators (objectives taken, gold advantage, and experience difference).

4) Confusion Matrices: Logistic Regression and SVM showed balanced precision and recall, effectively predicting both wins and losses. The Neural Network, however, had a high false-positive rate, indicating its tendency to incorrectly predict wins, highlighting potential overfitting to specific early-game scenarios.

5) Feature Importance: Logistic Regression coefficients emphasized the critical importance of dragons for both teams, aligning with the strategic value of securing early dragons in League of Legends due to the cumulative team buffs provided. Random Forest and XGBoost feature importance clearly highlighted gold and experience differences as the most predictive factors, reflecting the fundamental importance of early-game resource accumulation and management for eventual victory.

C. Interpretation of the Results

Across all evaluated models, Logistic Regression and Support Vector Machine consistently demonstrated the highest predictive performance, achieving strong ROC AUC scores (above 0.80) and balanced precision-recall tradeoffs. These models excelled at identifying meaningful patterns in early-game statistics—such as gold and experience differences, as well as early objective control (dragons and heralds)—which are known to be critical indicators of success in high-ranked League of Legends matches.

Tree-based models such as Random Forest and XGBoost performed competitively, with slightly lower ROC AUC scores, but still offered valuable insights through interpretable feature importance rankings. In contrast, the Multi-layer Perceptron underperformed in accuracy while retaining relatively high F1 scores, indicating potential overfitting or insufficient feature richness for deep learning models.

Visualization analyses reinforced these findings, highlighting the consistent importance of early-game economy and objectives. Specifically, dragon and gold differences were the most influential features, aligning with game mechanics where early advantages often snowball into victory.

IV. Conclusion

This study confirmed that early-game statistics offer strong predictive power for match outcomes in League of Legends, especially in high-ranked competitive settings. Features like gold and experience differentials, as well as early objective control (dragons and heralds), were consistently identified as the most influential, reinforcing the game's core snowball mechanics.

To further improve model accuracy and reflect the strategic complexity of competitive play, future enhancements may include:

1) Incorporating champion selection features: Integrating pick/ban phase data, including champion identities, synergies, and counter-picks, can provide essential strategic context and benefit complex models like neural networks.

2) Constructing interaction features: Modeling lane matchups, jungle pressure disparities, and inter-role synergies may better capture nonlinear influences on game outcomes.

3) Integrating temporal data: Segmenting the match into early/mid/late phases and incorporating evolving statistics could allow models to adapt to shifts in tempo and momentum.

Overall, this project demonstrates a viable data-driven framework for eSports win prediction and offers a foundation for more comprehensive and strategy-aware models in future work.

[1] Kaggle, “League of Legends Diamond Ranked Games (10 min),” [Online].Available:https://www.kaggle.com/datasets/bobbyscience/league-of-legends-diamond-ranked-games-10-min. [Accessed: Apr. 18, 2025].

[2] Riot Games Developer Portal, "Riot Games API Reference," [Online]. Available: https://developer.riotgames.com/apis. [Accessed: Apr. 18, 2025].

[3] G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, An Introduction to Statistical Learning with Applications in Python, 1st ed. New York: Springer, 2023.

[4] T. Hastie, R. Tibshirani, and J. Friedman, The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd ed. New York: Springer, 2009.